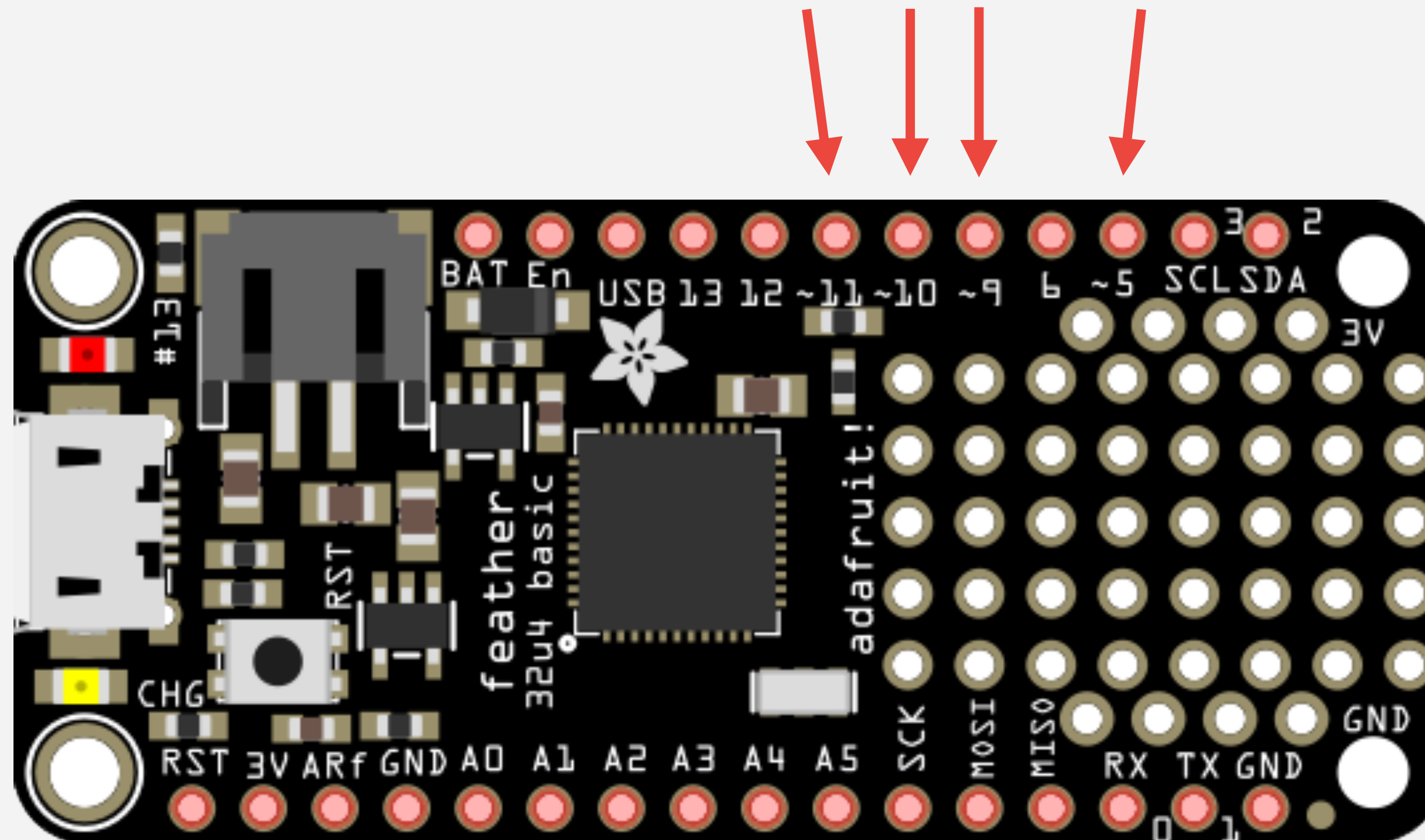


ANALOG OUTPUT AND BASIC AUDIO

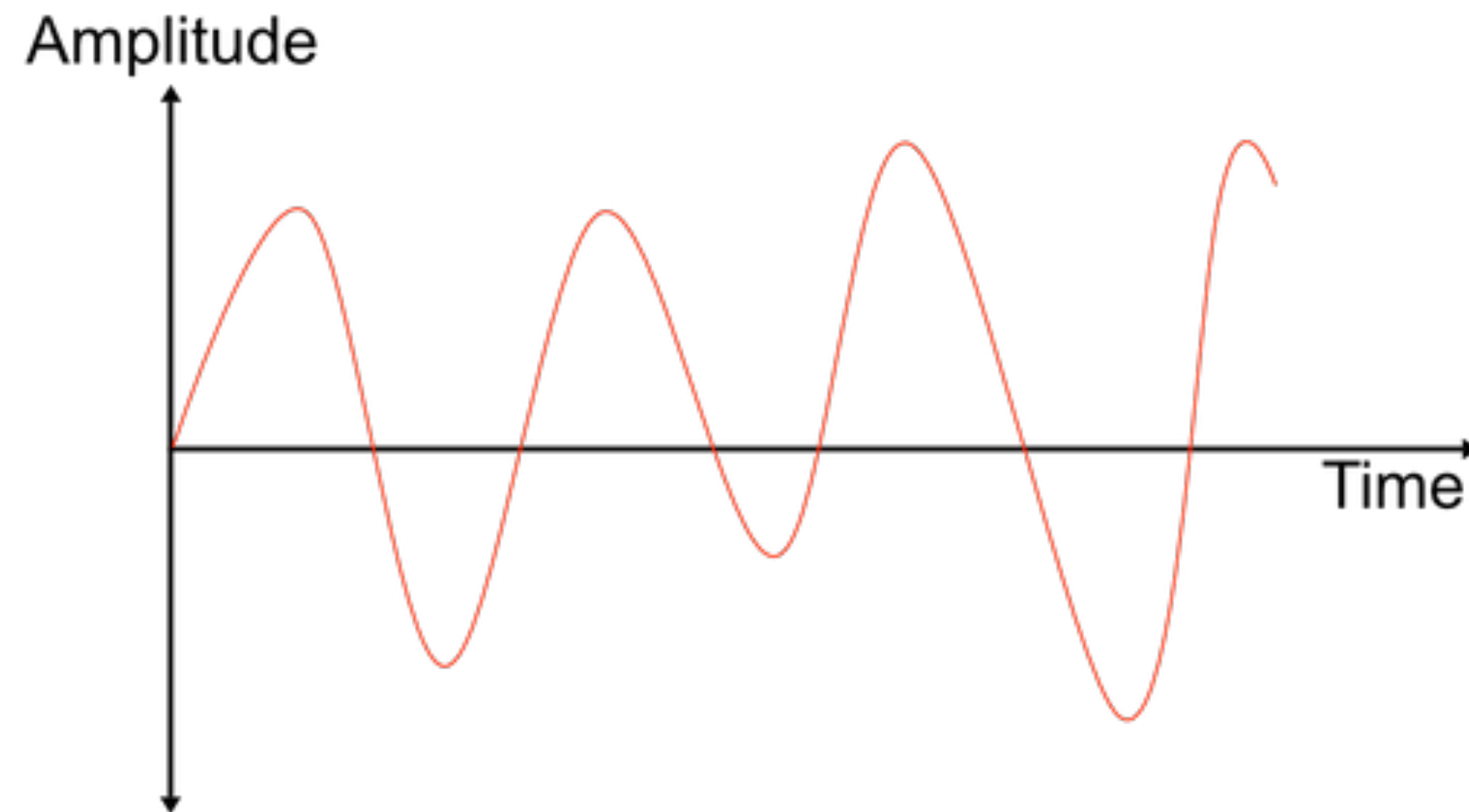
ANALOG OUTPUT



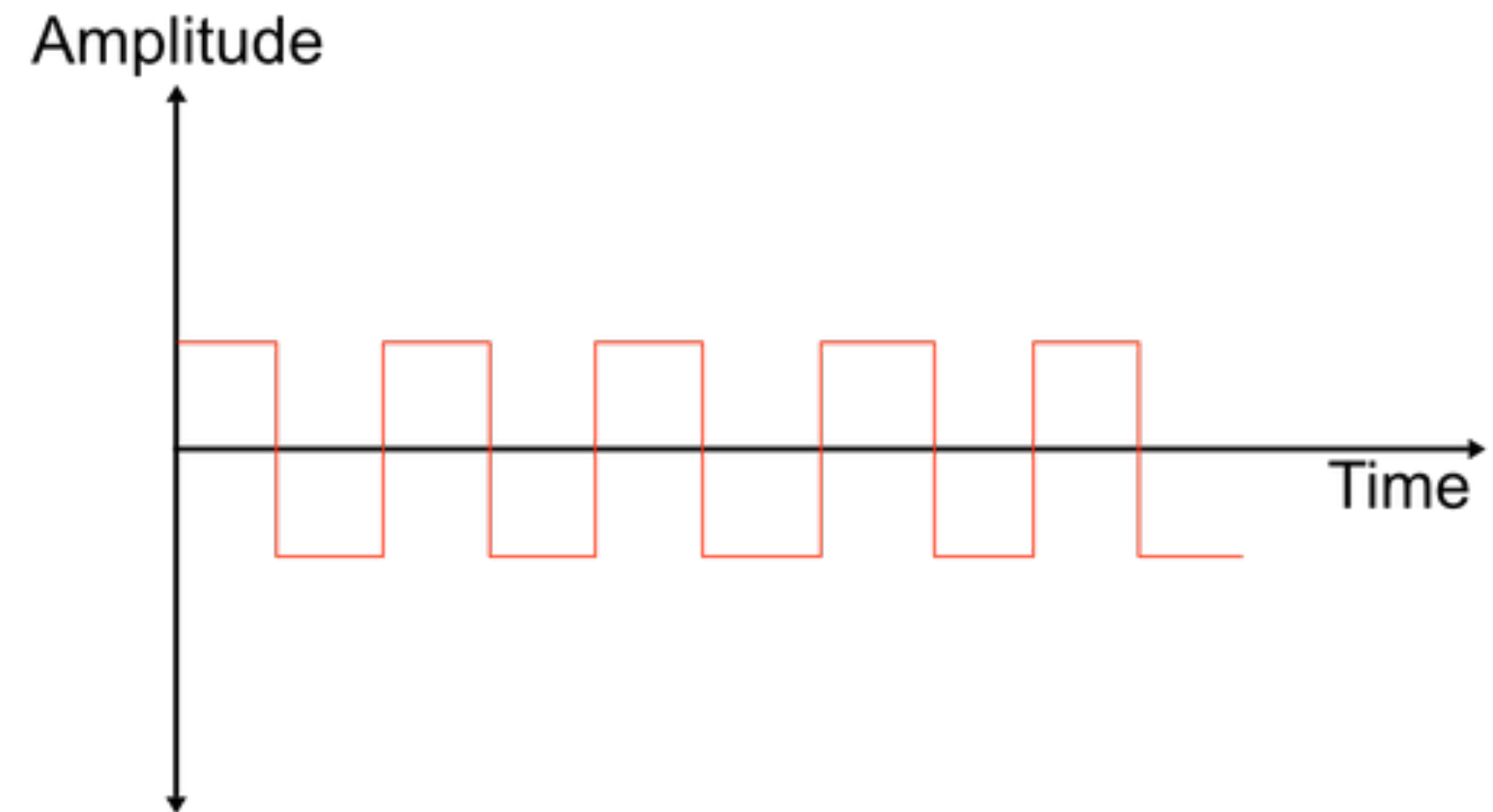
Analog Output*

*Only on the pins with the ~ next to them

Analog vs Digital



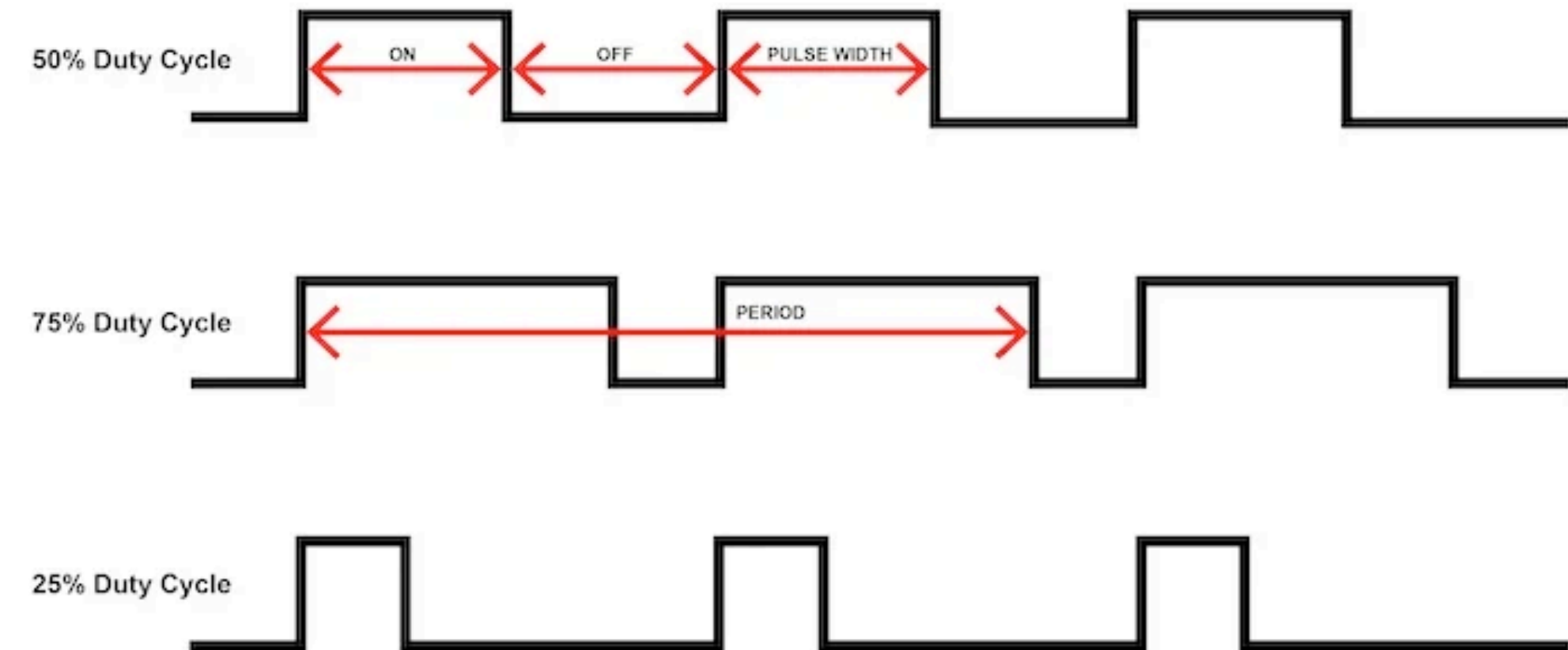
Analog Signal
Smooth gradient of
infinite values.



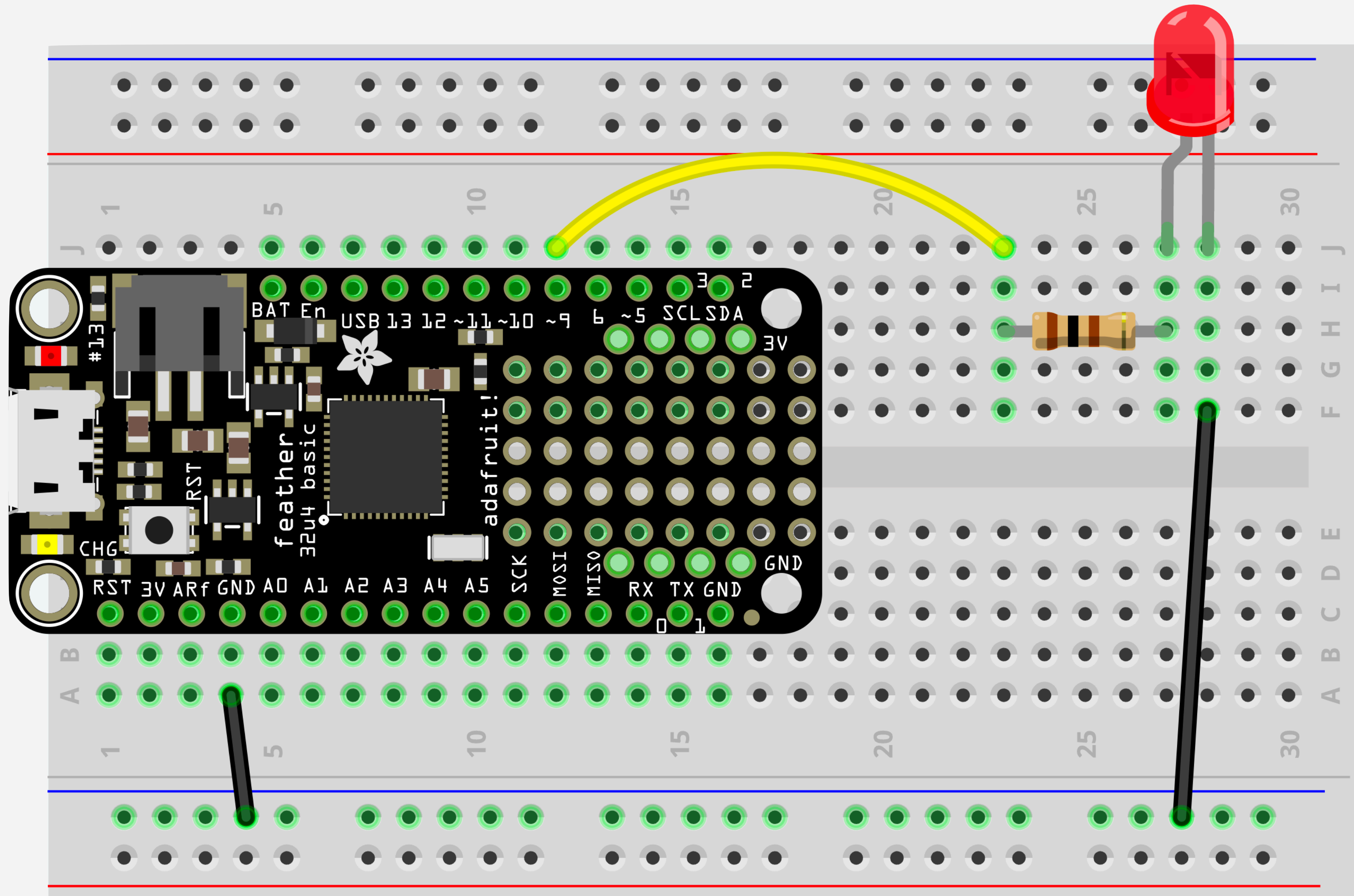
Digital Signal
Only 2 possible
values.

PWM – Pulse Width Modulation

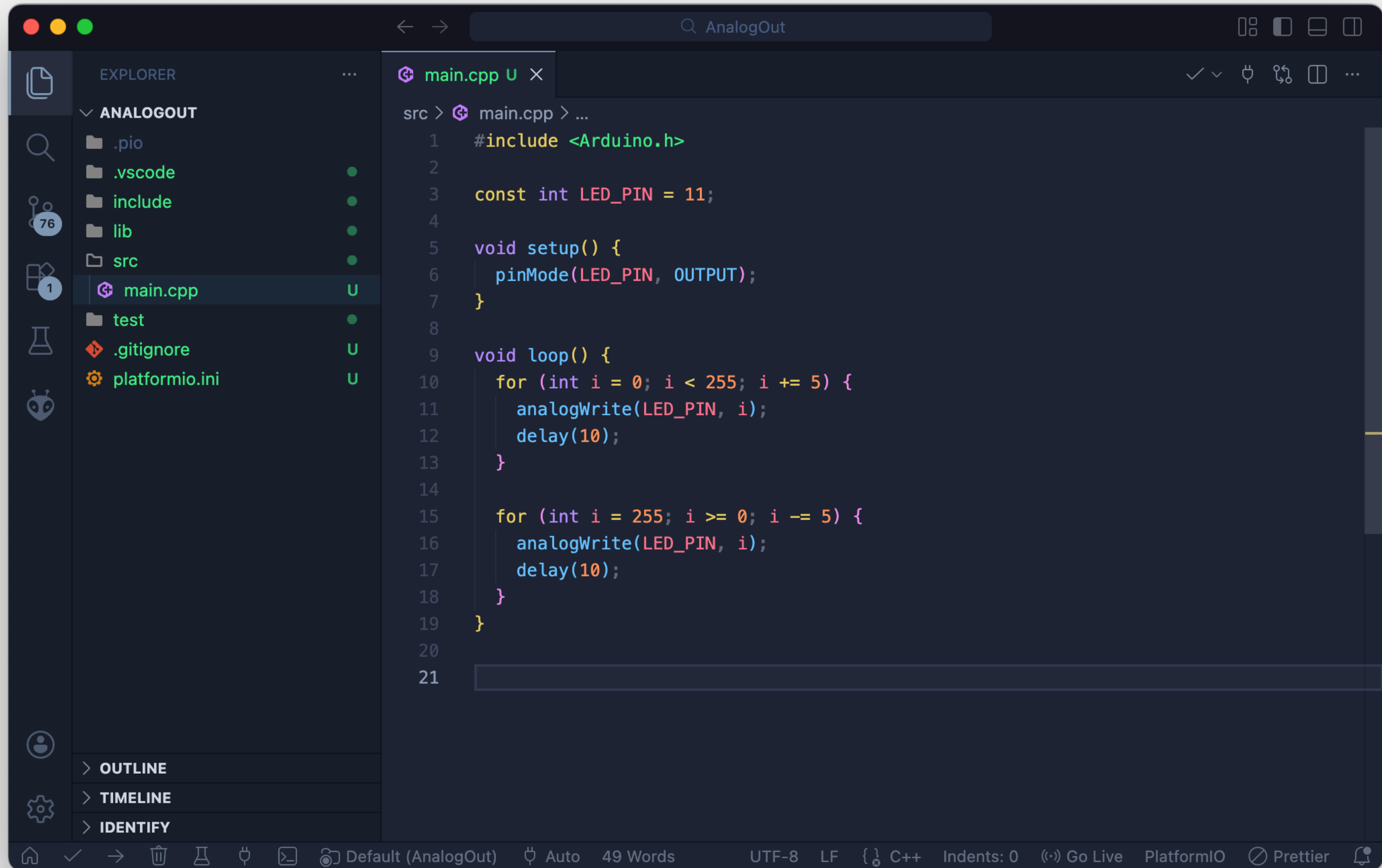
Most Arduino boards can't create true analog signals. We can however fake it, using a technique called PWM via the `analogWrite()` function.



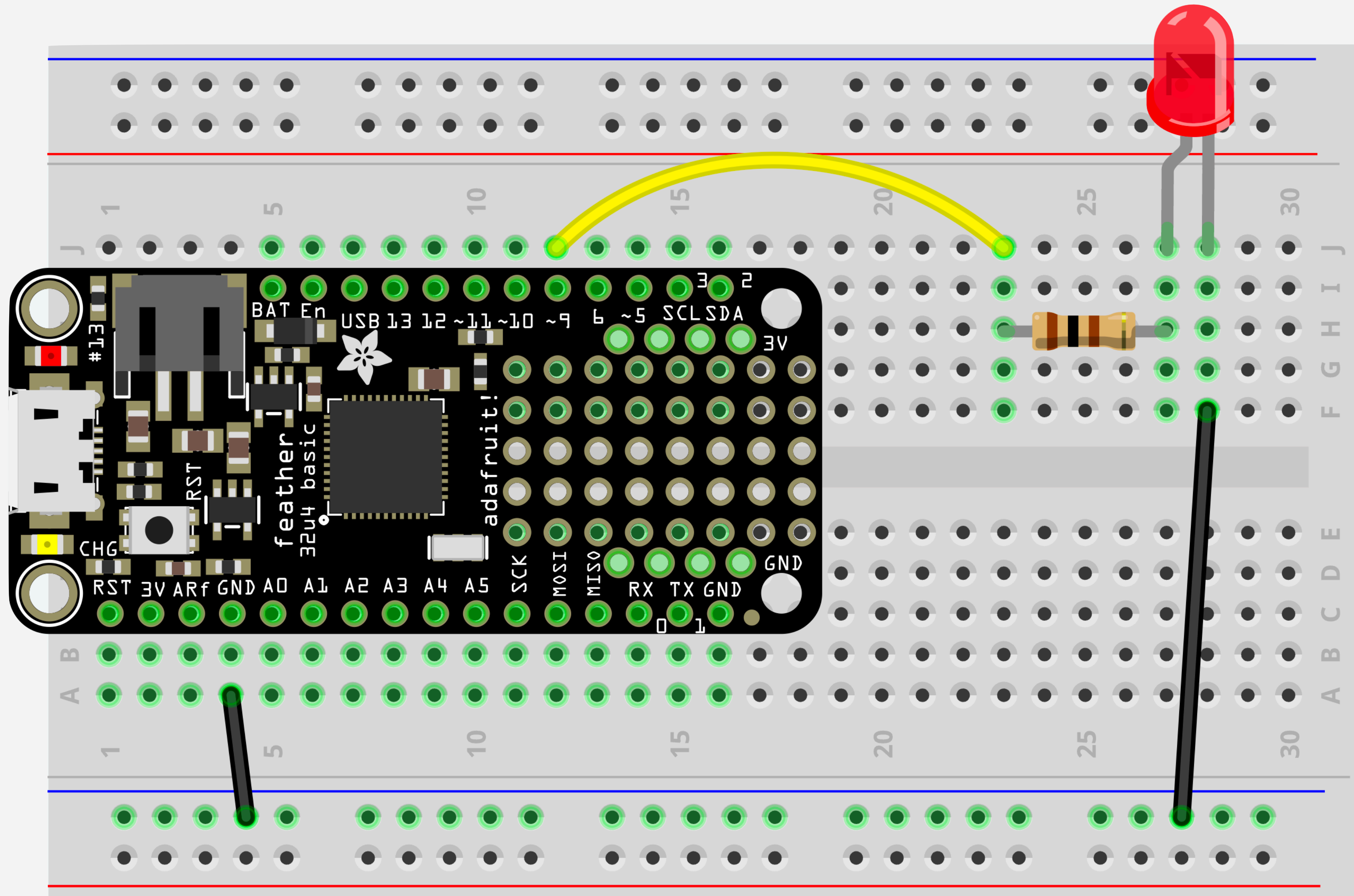
Fade a LED using a loop



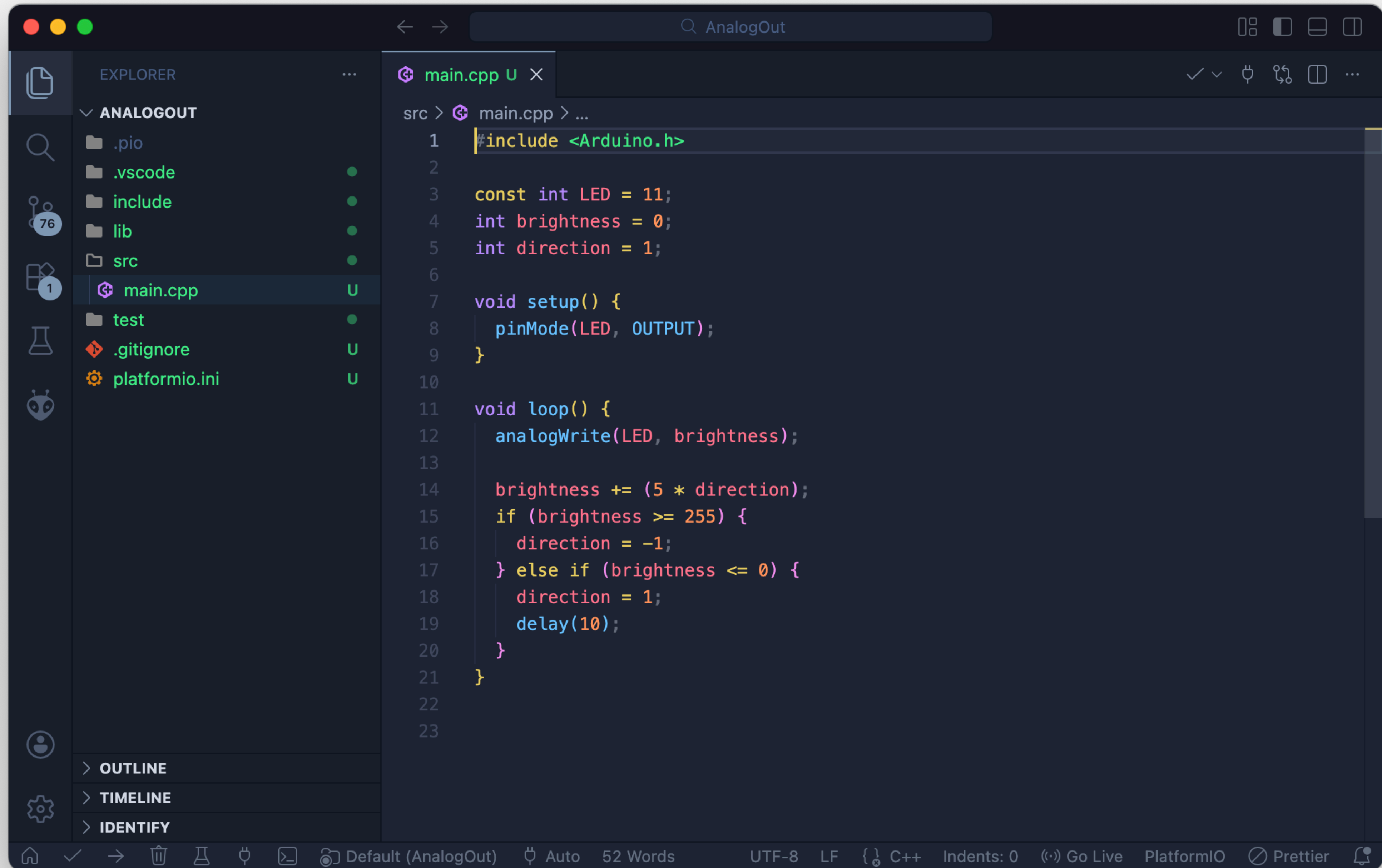
* Resistor is 100Ω (Brown Black Brown)



**Fade a LED
using a global variable**



* Resistor is 100Ω (Brown Black Brown)



BASIC AUDIO

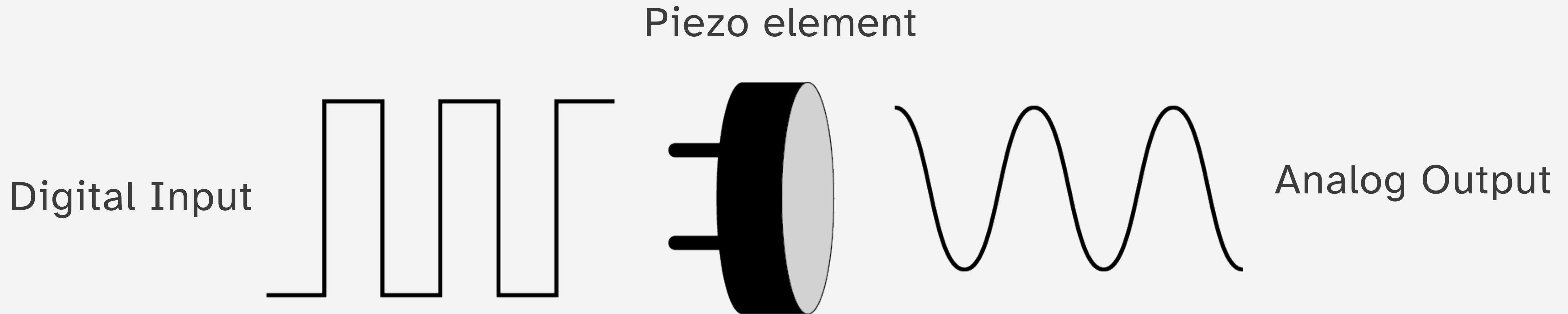
The Piezoelectric Effect

The piezoelectric effect is the ability of certain solid materials to generate an electrical charge when mechanical stress is applied.

This phenomenon involves the conversion of mechanical energy, such as pressure or vibration, into electrical energy.

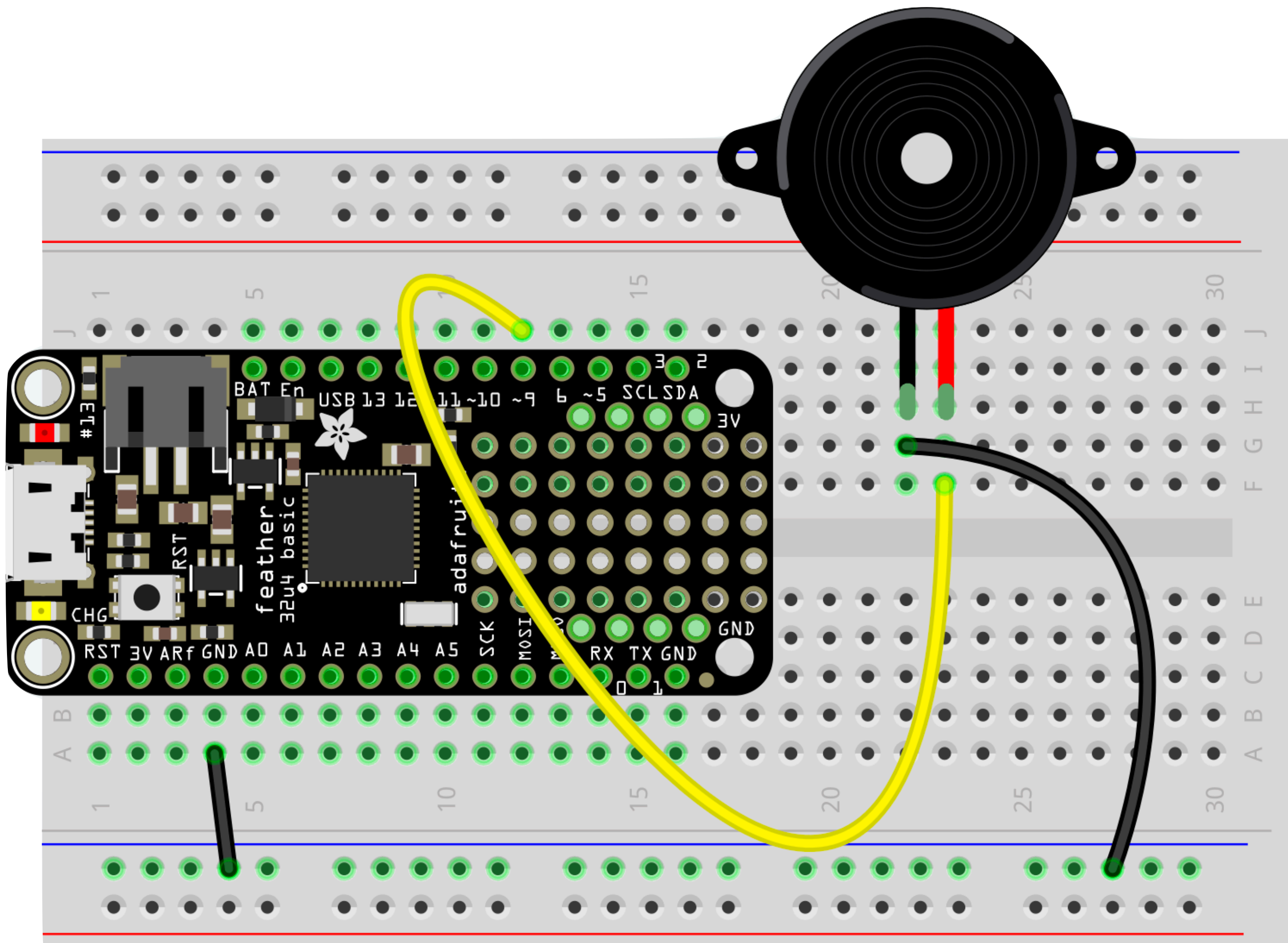
The effect also works in reverse, causing mechanical deformation when an electrical field is applied.

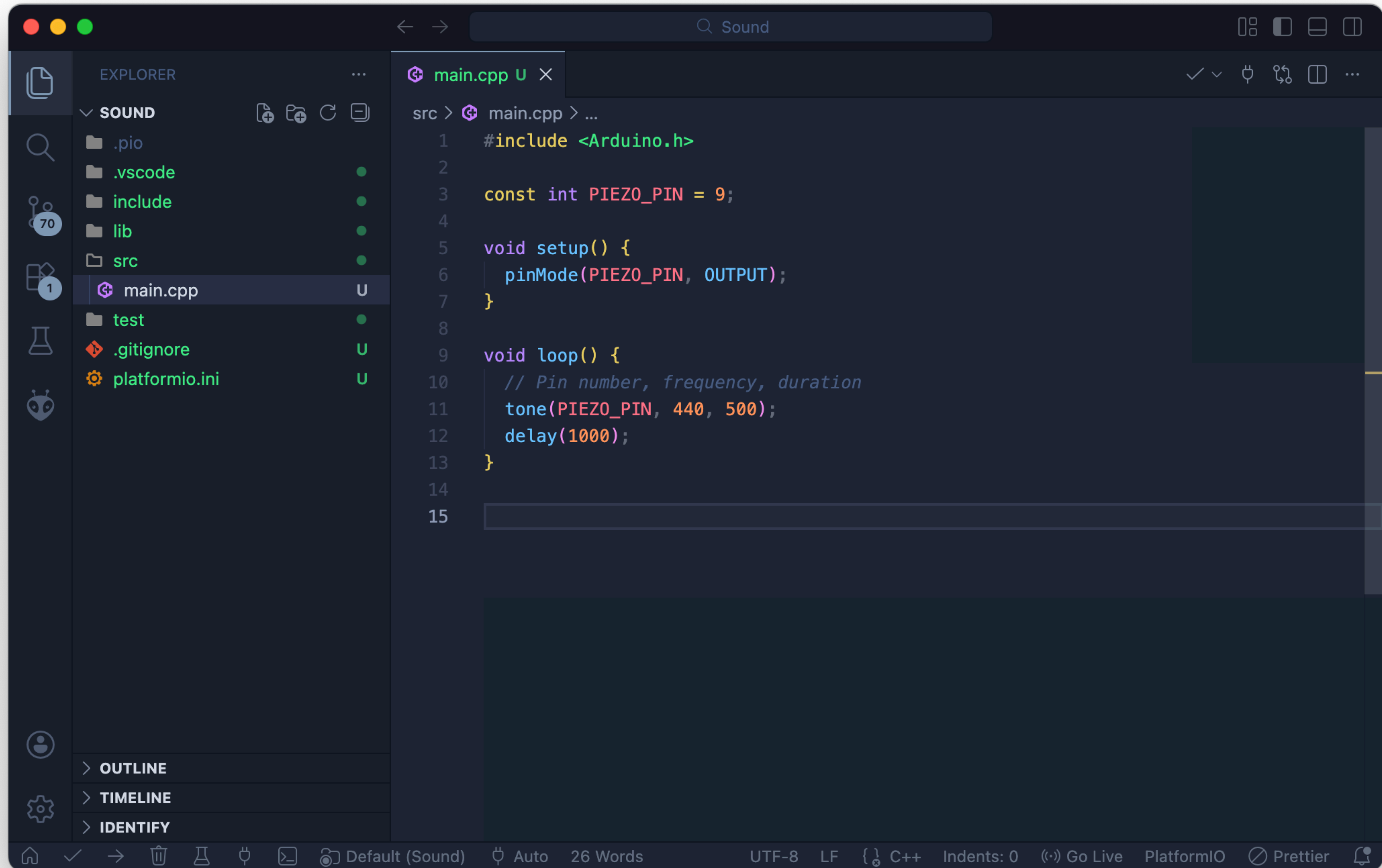
The Inverse Piezoelectric Effect



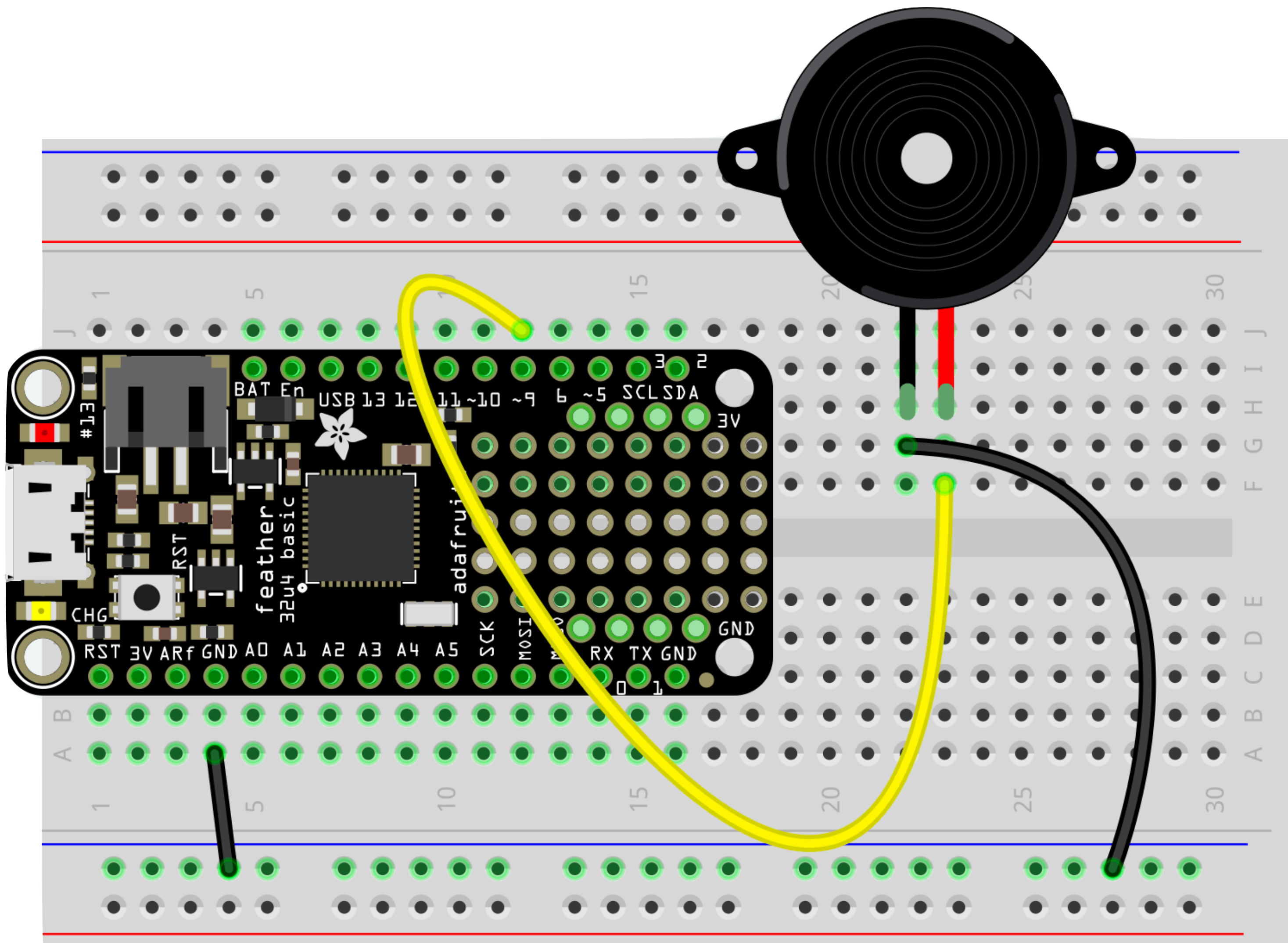
The piezoelectric effect also works in reverse, causing mechanical deformation when an electrical field is applied.

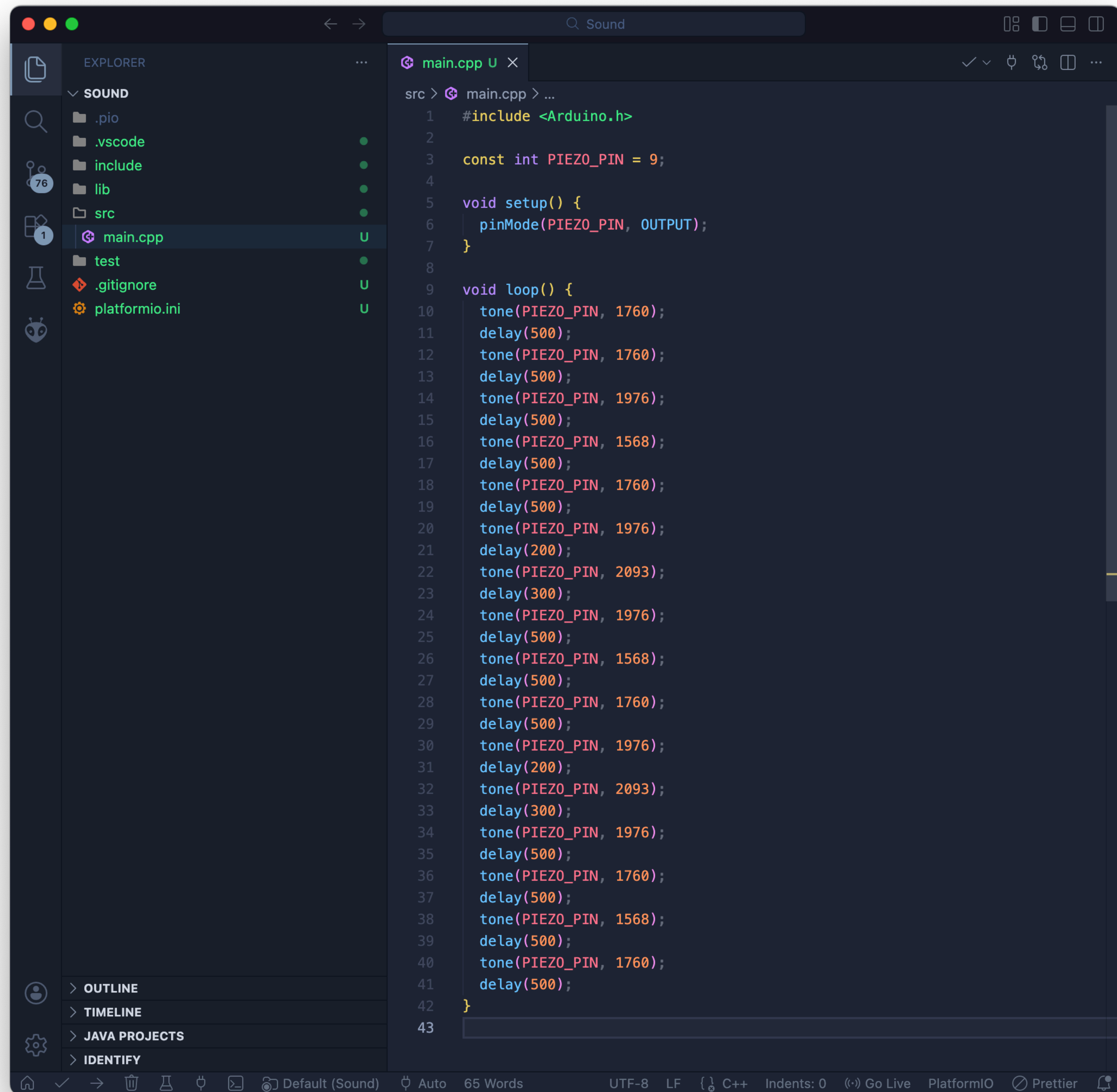
Play a single tone



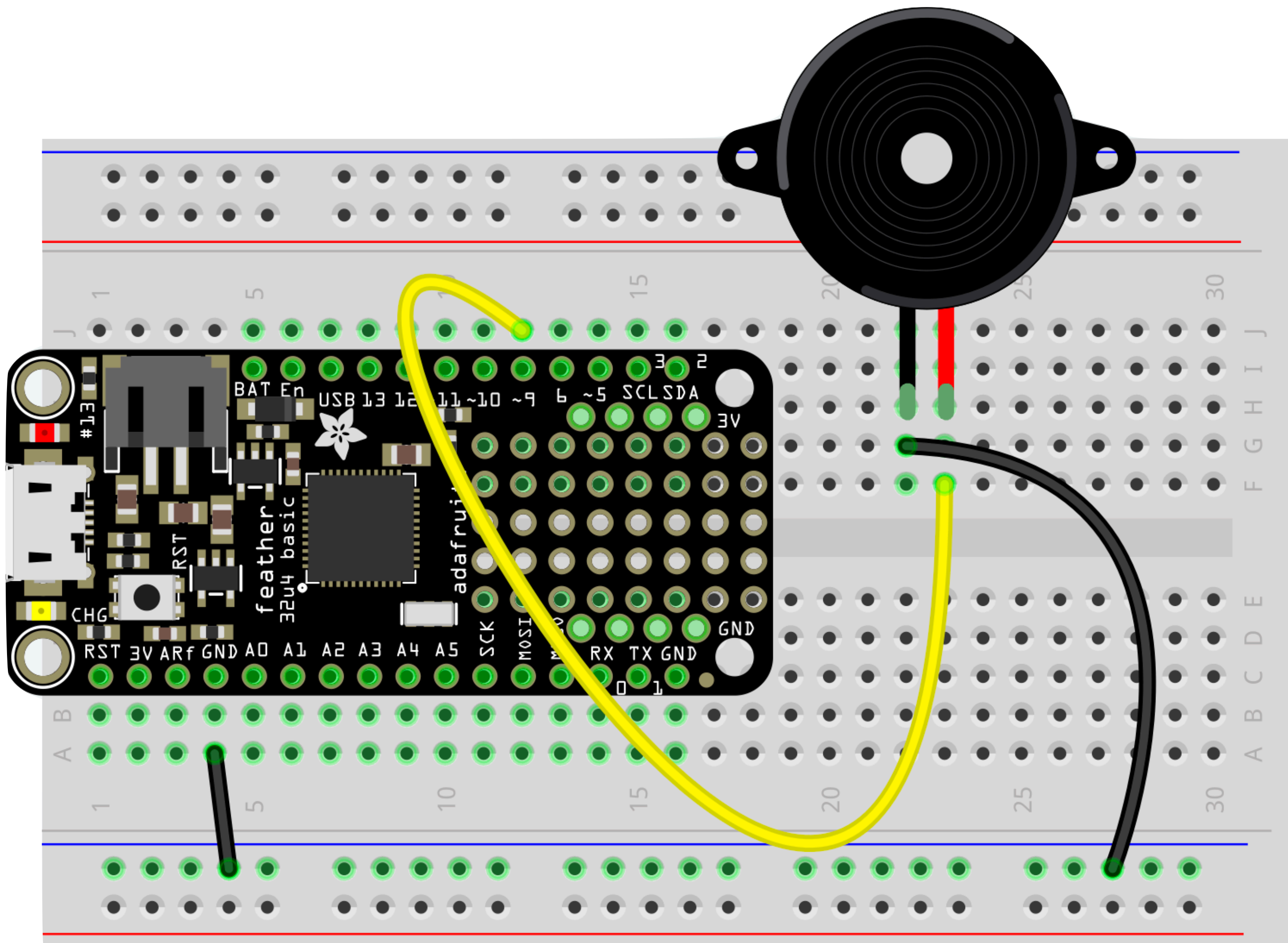


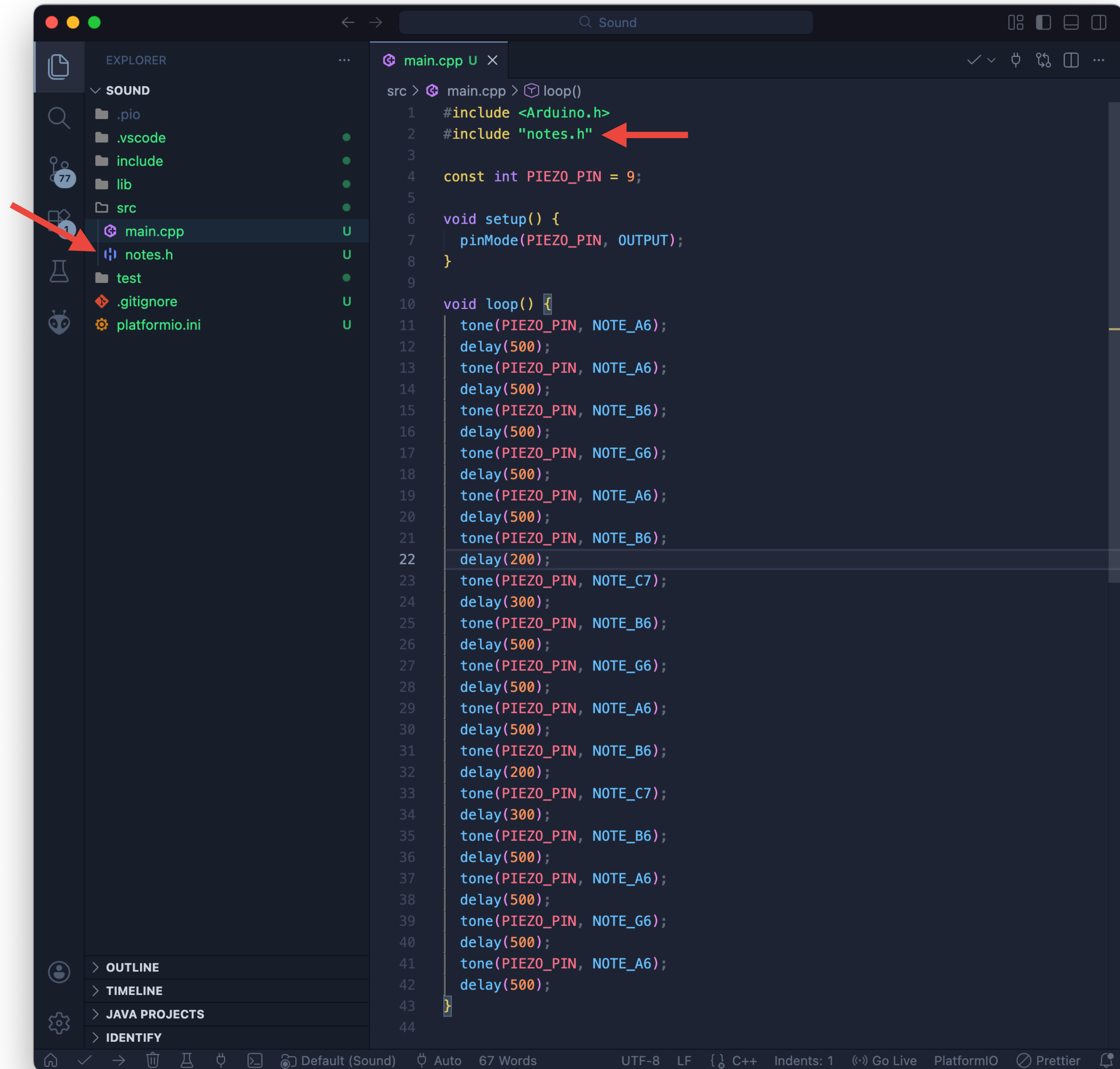
Play a song





**Play a song
using notes.h**





Try playing a specific song.